

1.

	PDMS		-
H ₂ O ₂	Stanford	Zare	
Contact Electrification		EDL	
1.		PDMS	PDMS
2.	10 ⁸ – 10 ⁹ V/m		
3.	ROS ORR	Grotthuss H ₂ O ₂	
		$\text{O}_2 \xrightarrow{\text{e}^-} \text{O}_2^{\bullet-} \xrightarrow{\text{H}^+} \text{HO}_2^{\bullet} \xrightarrow{\text{e}^-, \text{H}^+} \text{H}_2\text{O}_2$	
	topoHBNet	6 CP2K	AIMD

2.6

	H_2O_2 /	O_2 /	$HO_2^\bullet$	H_3O^+ /	(H- bonds)	Betti- 1 ()	(fs)	(LBHB) ¹
charged-surface-with-O2-interface (+O2)	0.3859	0.5621	0.6594	0.4281	151.92	56.19	136.10	0.002749
charged-surface-with-O2-water (+O2)	0.0003	1.0000	0.0003	1.0913	149.55	55.21	144.74	0.004277
charged-surface-without-O2 (O2)	0.0000	0.0000	0.0000	0.1270	157.37	61.12	158.03	0.003343
nofield ()	0.5697 ²	0.0000	0.0013	0.0417	154.80	61.66	229.33	0.001594
nofield-with-O2-interface (+O2)	0.0000	1.0000	0.0000	0.2826	157.92	63.58	200.58	0.002266
nofield-with-O2-water (+O2)	0.2170 ²	0.8087	0.2045	0.0685	148.11	54.84	188.02	0.001608

3. -

3.1 O_2

3.1.1 ORR

- charged-surface-with-O2-interface ($H_2O_2 = 0.3859$) charged-surface-with-O2-water ($H_2O_2 = 0.0003$) O_2 H_2O_2 O_2

						Grotthuss
	LBHB		$\text{HO}_2^\bullet$			
• LBHB			charged-surface-with-O2-water			LBHB
	0.004277	H_2O_2		0.0003	LBHB	
	H_2O_2		—	ORR	O_2	
				O_2		-
	PCET	H_2O_2				

Footnotes

- LBHB

 mean_lbhb_fraction

 ~3-18%

 proton_transfer_analysis.py → proton_dynamics_summary.json

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 aligned_topology_transport.csv


 LBHB
- H_2O_2

 ORR

 $\text{O}_2 \rightarrow \text{H}_2\text{O}_2$

 $2\text{H}_2\text{O} \rightarrow 2^\bullet\text{OH} + \text{H}_2 \rightarrow \text{H}_2\text{O}_2 + \text{H}_2$

 §3.1.2



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 AIMD

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